

KOGI PLATFORM

Marketplace & Exchange

System Design Document

v1.0 · Kogi Platform · March 2026

kogi-marketplace · kogi-exchange

*The economic layer of the Kogi Independent Worker Operating System —
where workers trade skills, assets, labor, and financial instruments.*

Kogi Team · Independent Worker Operating System · Confidential

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1. OVERVIEW

1.1 Purpose & Vision

The Kogi Marketplace and Exchange constitute the economic layer of the Kogi Independent Worker Operating System. Together they form the transactional, liquidity, and matching infrastructure that connects independent workers, collectives, cooperatives, and organizations to one another for the purpose of trading labor, skills, assets, resources, financial instruments, and portfolio components.

Where kogi-office organizes the worker's private portfolio, and kogi-community connects workers socially, the Marketplace and Exchange are the surfaces through which portfolio value is realized: skills are hired, assets are sold, deals are struck, capital is raised, and portfolio components are exchanged.

Design Principle: The Marketplace is where you offer what you have. The Exchange is where you trade what it's worth. Both are powered by the same Portfolio abstraction — every listing, bid, offer, and deal is ultimately a Portfolio Item with economic intent.

1.2 Position in the Kogi Platform

The Marketplace and Exchange sit at the boundary between a worker's private operating world and the broader Kogi network economy. Community feeds discovery into both; the Portfolio System is the source of all tradable entities; the Bank provides payment infrastructure; and the Engine powers matching, pricing, and risk intelligence.

Layer	System	Relationship
Upstream	kogi-office / Portfolio System	Source of all listings — Assets, Resources, Artifacts with visibility=Public
Upstream	kogi-community	Discovery engine — posts, spaces, and social graph feed deal flow
Downstream	kogi-bank	Payment, escrow, invoicing, wallet, and settlement layer
Downstream	kogi-engine	Matching, recommendation, risk scoring, and price intelligence
Peer	kogi-gig	Work contracts and scheduled engagements originate from Exchange deals
Peer	kogi-organizations	Cooperatives, collectives, and DAOs operate campaigns and group buys

External	3rd Party Integrations	Upwork, Fiverr, Behance, Robinhood, Coinbase, SoFi, and others
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1.3 Module Codes

Module	Code	Service	Language
Marketplace	KMKT	kogi-marketplace	Go (service) · Rust (core)
Exchange	KEXC	kogi-exchange	Go (service) · Rust (core)
Deal Engine	KDEAL	kogi-deal	Go (service)
Escrow / Settlement	KESC	kogi-escrow (via kogi-bank)	Go (service)

2. kogi-marketplace

2.1 Concept

kogi-marketplace is the platform's primary buying-and-selling surface. It enables independent workers and organizations to list, discover, acquire, and sell skills, labor, assets, resources, artifacts, templates, playbooks, and any other portfolio-derived value. It is simultaneously a labor market, an asset store, a service marketplace, and a crowdfunding platform — unified under one coherent UI and data model.

Every item listed on the Marketplace is directly connected to a Portfolio Component. This means every listing carries the full context of its creator's portfolio: version history, analytics, governance metadata, and AI-computed health and maturity scores — giving buyers unprecedented transparency and sellers differentiated, trust-backed offerings.

2.2 Marketplace Categories

The Marketplace is organized across four primary categories:

Category	Description	Examples	Portfolio Link
Work	Labor, skills, and services offered for hire	Gigs, contracts, jobs, tasks, consulting	Resource (Human) Items
Assets & Artifacts	Purchasable or licensable portfolio outputs	Playbooks, templates, guides, tools, designs, software	Asset / Artifact Items
Resources	Allocatable resources offered to others	Compute, licenses, equipment, capital	Resource Items
Campaigns	Fundraising, crowdfunding, and group economics	Equity rounds, resource pools, community funds	Program / Project Items

2.3 Listings

2.3.1 Listing Data Model

Every Marketplace listing is a Portfolio Item of type Asset, Resource, or Artifact with visibility=Public, extended with the following marketplace-specific fields:

Field	Description
listing.id	Platform-unique UUID
listing.type	work asset artifact resource campaign template service bundle

listing.portfolio_component_id	The source Portfolio Component ID (Asset / Resource / Artifact)
listing.owner_id	Listing creator's user ID
listing.title	Display title of the listing
listing.description	Rich-text description (markdown-compatible)
listing.price	Price struct: { amount, currency, pricing_model }
listing.pricing_model	fixed hourly daily subscription negotiable free auction revenue_share
listing.status	draft active paused sold expired archived
listing.visibility	public community invite_only
listing.tags[]	Topic and skill tags for discovery
listing.skills[]	Required or offered skills (for work listings)
listing.media[]	Images, videos, portfolio previews
listing.reviews[]	Linked Review entity IDs
listing.rating	Aggregate rating (0–5, computed)
listing.orders[]	Linked Order entity IDs
listing.proposals[]	Linked Proposal entity IDs
listing.analytics	Impressions, CTR, conversion rate, views, engagement
listing.ai_score	AI-computed quality/match score from kogi-engine
listing.metadata	Full ComponentMetadata: id, owners, tags, policy_ids, timestamps, version

2.3.2 Listing Lifecycle

Listings follow a governed lifecycle managed by the Portfolio System's ApprovalWorkflow and status machine:

State	Description	Transitions
Draft	Private, under construction. Not visible to others.	→ Active (publish), → Archived
Active	Publicly discoverable and available for purchase/hire.	→ Paused, → Sold, → Expired, → Archived
Paused	Temporarily hidden by owner.	→ Active, → Archived
Sold	Work or asset fully contracted/purchased. Fulfilled.	→ Active (re-list), → Archived
Expired	Past end date or automated expiry.	→ Active (renew), → Archived

Archived	Deep storage. Recoverable.	→ Active (restore)
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2.4 Work Marketplace (Labor Market)

The Work sub-market is kogi-marketplace's labor exchange — connecting workers who offer skills, services, and time with those who need them. It covers the full spectrum from short-term gigs to long-term contracts, one-off tasks to ongoing employment relationships.

2.4.1 Work Listing Types

Type	Description	Duration	Engagement Model
Gig	Single, defined deliverable task	Hours – Days	Fixed price or hourly
Task	Discrete sub-work unit within a larger project	Hours	Fixed price
Contract	Ongoing service engagement under agreed terms	Weeks – Months	Retainer / milestone
Job	Longer-term role or embedded position	Months+	Salary / daily rate
Consulting	Advice, strategy, or expert review	Variable	Hourly / project
Collaboration	Joint creative or technical project partnership	Variable	Revenue share / co-ownership

2.4.2 Worker Profile & Skill Graph

Each user's work offering is backed by their Portfolio Resource Item (type=Human), which contains:

- `skills[]` — tagged skill taxonomy (design, engineering, writing, legal, finance, etc.)
- `availability` — schedule windows pulled from kogi-gig ScheduleBook
- `rate` — hourly/daily/project rates
- `portfolio_links[]` — linked public Portfolio Components as proof of work
- `reviews[]` — aggregated reviews from completed engagements
- `certifications[]` — credential badges and verified qualifications
- `ai_reputation_score` — kogi-engine computed trust score based on completion rate, review sentiment, and portfolio health

2.5 Asset & Artifact Marketplace

Workers can publish and sell the outputs of their portfolio — documents, playbooks, templates, designs, software, datasets, research, and any other artifact or asset — as purchasable or

licensable Marketplace listings. This is how knowledge, creative work, and intellectual products become economic assets on the platform.

2.5.1 Asset/Artifact Listing Types

Type	Examples	Pricing Models
Playbook	Investment playbooks, process guides, strategy docs	Fixed, Subscription, Free
Template	Project templates, proposal templates, contract starters	Fixed, Free
Design / Blueprint	UI kits, architecture diagrams, brand kits	Fixed, License
Software / Tool	Scripts, plugins, automation workflows, SDK extensions	Fixed, Subscription, License
Dataset / Research	Market research, curated data sets, analytics reports	Fixed, Subscription
Course / Guide	Instructional content, how-to guides, tutorials	Fixed, Subscription
Contract Template	Legal templates, NDAs, service agreements	Fixed, Free
Bundle	Curated collection of multiple assets	Fixed, Subscription

Example: A user creates a Real Estate Investment Playbook as a Book item in their Portfolio. They publish it to the Marketplace at a fixed price. The kogi-engine RecommendationEngine surfaces it to users who have created Real Estate project types. Subscribers receive automatic updates when a new version is published. This is the end-to-end flow from portfolio to market to subscription.

2.6 Proposals & Deals

2.6.1 The Proposal Flow

The primary transactional mechanism in the Marketplace is the Proposal → Deal flow. Rather than simple one-click purchases, most Marketplace engagements involve a structured negotiation that preserves context, terms, and documentation through the full lifecycle.

Stage	Description
Inquiry	Buyer sends initial message or inquiry to a listing. Creates a Deal Room.
Proposal	Seller or buyer submits a formal Proposal: scope, deliverables, timeline, price, terms.

Negotiation	Counter-proposals exchanged in the Deal Room. AI assistant can draft and compare terms.
Accepted	Both parties approve. Proposal transitions to a Deal.
Escrow Funded	Buyer funds the escrow via kogi-bank. Work begins.
Milestone Check	Milestone deliverables reviewed. Partial escrow releases triggered.
Completion	Final deliverable accepted. Full escrow released.
Review	Both parties submit ratings and reviews. Portfolio analytics updated.

2.6.2 Proposal Data Model

Field	Description
proposal.id	UUID
proposal.listing_id	Source listing reference
proposal.buyer_id / seller_id	Party references
proposal.status	draft sent countered accepted rejected expired cancelled
proposal.scope	Rich text: deliverables, exclusions, assumptions
proposal.timeline	Start date, end date, milestone schedule
proposal.price	Proposed price with pricing_model
proposal.milestones[]	Milestone items: name, due_date, amount, deliverable_description
proposal.terms	Payment terms, IP assignment, revision policy, cancellation policy
proposal.attachments[]	Supporting files, portfolio component links
proposal.deal_room_id	Linked Deal Room ID
proposal.metadata	Full ComponentMetadata

2.7 Campaigns, Crowdfunding & Group Economics

kogi-marketplace supports collective economic activity through Campaigns — a mechanism for resource gathering, crowdfunding, cooperative purchasing, and equity participation. Campaigns are closely integrated with kogi-organizations and kogi-bank.

2.7.1 Campaign Types

Campaign Type	Description	Primary Actor
Resource Campaign	Collective gathering of tools, compute, or materials	Collective / Individual

Fundraise	Donation-based funding for a project or mission	Individual / Collective
Equity Crowdfunding	Fractional ownership stake in a portfolio project or org	Organization / Cooperative
Group Buy	Collective purchasing of an asset or license at bulk rate	Community / Space
Revenue Share Pool	Pool of backers sharing in a project's future revenue	Organization
Community Fund	Ongoing mutual-aid or community support pool	Cooperative / Federation
Pre-sale	Early access or pre-purchase of an asset before launch	Individual / Org

2.7.2 Campaign Data Model

Field	Description
campaign.id	UUID
campaign.type	resource fundraise equity group_buy revenue_share community_fund presale
campaign.portfolio_component_id	Linked Portfolio Program or Project being funded
campaign.goal	Funding goal: { amount, currency, resource_units }
campaign.raised	Current amount raised
campaign.backers[]	User IDs and contribution amounts
campaign.equity_terms	For equity campaigns: % offered, valuation, rights, vesting schedule
campaign.deadline	Campaign end date
campaign.status	draft active funded failed distributing closed
campaign.milestones[]	Use-of-funds milestones with release conditions
campaign.org_id	Optional linked Organization entity

2.8 Actions & Interactions

The full set of user-facing actions available on Marketplace entities:

Action	Actor	Effect	Portfolio Update
List	Seller	Creates a Listing from a Portfolio Component	Sets visibility=Public

Browse / Search	Buyer	Discovery via SearchEngine + RecommendationEngine	No change
Save / Bookmark	Buyer	Saves listing to collection	Bookmark count +1
Inquire	Buyer	Opens a Deal Room and sends initial inquiry	Creates Deal Room
Propose	Buyer / Seller	Submits formal Proposal	Creates Proposal entity
Accept	Counterparty	Approves Proposal → Deal	Status → Active Deal
Purchase	Buyer	Direct buy of fixed-price asset	Order created
Fund Escrow	Buyer	Deposits payment into escrow via kogi-bank	Escrow funded
Deliver	Seller	Marks deliverable as submitted	Milestone update
Approve Delivery	Buyer	Accepts deliverable, triggers escrow release	Escrow release
Rate & Review	Both	Submits rating and review after completion	Review analytics updated
Invest	Investor	Commits to equity campaign contribution	ActionKind::Invest
Donate	Donor	Contributes to a fundraise campaign	ActionKind::Donate
Subscribe	User	Subscribes to a seller's ongoing content or updates	Subscriber count +1
Report	User	Flags listing for review	Moderation queue
Share	User	Shares listing to Community feed or externally	Share count +1

2.9 Third-Party Integrations

Marketplace tiles can link to and import from external platforms, managed by the kogi-providers ProviderSystem:

Category	Platforms	Integration Type
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Freelance Markets	Upwork, Fiverr, Toptal, Freelancer	Profile import, listing sync, order bridging
Portfolio / Creative	Behance, Dribbble, GitHub, GitLab	Portfolio import, project showcase sync
Job Markets	LinkedIn, Indeed, AngelList	Job listing sync, profile export
Collaboration	Slack, Notion, Google Workspace, Microsoft Teams	Project room linking, file sync
Commerce	Gumroad, Shopify, Stripe	Asset sales routing, payment bridging

3. kogi-exchange

3.1 Concept

kogi-exchange is the platform's structured trading and value-exchange layer. Where the Marketplace facilitates buying and selling, the Exchange facilitates bidding, offering, trading, and structured deal-making across a broader set of value types: financial instruments, portfolio commodities, equity positions, labor contracts, and resource allocations.

The Exchange supports order-book style mechanics (bids and offers), negotiated deals, and peer-to-peer swaps — all within a governed, auditable framework connected to the Portfolio System. It is designed to support both the day-to-day operational needs of independent workers (e.g., trading a contract slot) and more sophisticated financial activity (e.g., equity crowdfunding rounds, portfolio asset sales, resource swaps).

Design Principle: The Exchange is not just a financial market — it is a multi-dimensional portfolio economy. Labor, knowledge, capital, resources, and ownership stakes are all first-class tradable entities in the Kogi Exchange.

3.2 Exchange Markets

kogi-exchange is organized across four distinct but integrated markets:

Market	Code	What is Traded	Mechanics
Labor Market	KLM	Skills, gigs, contracts, work capacity, tasks	Bids & offers, proposals, matching
Portfolio Commodities Market	KPCM	Portfolio Items: assets, artifacts, projects, programs	Listings, offers, deals, swaps
Financial Instruments Market	KFIM	Equity, revenue share, convertible notes, funds, tokens	Order book, auctions, rounds
Resource Market	KRM	Compute, licenses, physical assets, capital, IP rights	Spot trades, futures, leases

3.3 Order Book & Trading Mechanics

3.3.1 Bids, Offers & Orders

The Exchange implements a structured order system supporting multiple execution models:

Order Type	Description
Bid	A buyer's offer to purchase at a specified price or terms. Remains open until matched, expired, or cancelled.
Ask / Offer	A seller's offer to sell at a specified price or terms.
Market Order	Immediate execution at the best available price.
Limit Order	Execute only at a specific price or better.
Proposal	Structured negotiated offer — full terms, scope, timeline, price.
RFP (Request for Proposal)	Buyer posts requirements; sellers submit competing proposals.
Auction	Time-limited competitive bidding; highest bid wins.
Reverse Auction	Buyer posts a budget ceiling; sellers compete on price and quality.
Swap	Direct peer-to-peer exchange of portfolio items, resources, or labor — no currency required.
OTC Deal	Over-the-counter bilateral negotiated deal, off order book.

3.3.2 Order Data Model

Field	Description
order.id	UUID
order.market	labor commodities financial resource
order.type	bid ask market limit proposal rfp auction swap otc
order.subject	The Portfolio Component, financial instrument, or resource being traded
order.price	{ amount, currency, pricing_model, min_price (for auctions) }
order.quantity	Units, hours, shares, licenses, etc.
order.terms	Delivery terms, conditions, expiry
order.status	open matched pending_escrow in_progress completed cancelled expired
order.counterparty_id	Matched counterparty once paired
order.deal_id	Linked Deal entity upon match
order.deal_room_id	Linked Deal Room for negotiation
order.metadata	Full ComponentMetadata with audit trail

3.4 Labor Market

The Labor Market is the Exchange's primary operational market for independent workers. It supports the full lifecycle of finding, hiring, contracting, executing, and settling work engagements.

3.4.1 Labor Market Flows

Flow	Initiator	Description
Post & Apply	Buyer posts requirement; Workers apply	Standard job/gig posting with application flow
Search & Propose	Worker searches listings; Proposes terms	Outbound worker-driven deal initiation
AI Match & Introduce	kogi-engine initiates	MatchEngine surfaces worker-buyer pairs proactively
RFP	Buyer posts structured requirements	Multiple workers submit competing proposals
Direct Hire	Buyer invites specific worker	Targeted engagement without open listing
Collective Staffing	Organization posts team requirement	Fills multiple roles via a single coordinated campaign

3.4.2 Labor Contract Types

Contract Type	Description	Payment Model	Portfolio Integration
Gig Contract	Single fixed deliverable	Fixed + Escrow	Creates Project item
Retainer	Ongoing access to worker capacity	Recurring billing via kogi-bank	Creates Resource allocation
Milestone Contract	Multi-phase project with staged payments	Escrow per milestone	Links to sprint milestones
Revenue Share Contract	Worker paid as % of project revenue	Ongoing via kogi-bank trigger	Links to Asset revenue tracking
Equity Contract	Worker compensated with ownership stake	Token / share issuance	Links to Organization cap table
Barter / Swap	Skill or service traded for another skill or asset	No currency — swap settlement	Dual Portfolio updates

3.5 Portfolio Commodities Market

The Portfolio Commodities Market enables direct trading of Portfolio Items — projects, programs, assets, artifacts, and other components. This is where portfolio value becomes directly tradable: a completed project can be sold, a program portfolio can be acquired, an artifact collection can be licensed.

3.5.1 Tradable Portfolio Item Types

Item Type	Trade Mechanism	Valuation Model
Asset	Direct sale, license, auction	AssetValue computational model from kogi-engine
Artifact	License, sale, subscription	ArtifactMaturity score + market comps
Project (Completed)	Acquisition, IP transfer	ProjectMetrics model + negotiated valuation
Program	Partial stake, full acquisition	ProgramAlignment + SubPortfolioRollup models
Resource (Machine / License)	Rental, lease, sale	ResourceUtilisation model
Template / Playbook	Fixed sale, subscription	Engagement analytics + follower count
Tool / Integration	License, subscription, sale	Usage analytics + TMS toolbox data

3.6 Financial Instruments Market

The Financial Instruments Market is kogi-exchange's most sophisticated sub-market. It supports structured financial transactions including equity sales, convertible instruments, revenue participation, and token-based arrangements — all anchored to real portfolio entities.

3.6.1 Supported Instruments

Instrument	Description	Settlement
Equity Stake	Fractional ownership in a project, program, or organization	Cap table update via kogi-organizations
Revenue Share Note	Right to a % of future revenue from a portfolio component	Automated trigger via kogi-bank revenue tracking
Convertible Note	Debt that converts to equity upon a trigger event	Conversion event tracked in PortfolioSystem
Community Bond	Debt instrument issued by a cooperative or collective	Fixed-term repayment via kogi-bank
Donation / Grant	Non-repayable capital contribution to a mission	Donor records via ActionKind::Donate
Token / Digital Asset	Platform-native or external blockchain asset	Bridge to external chains (Ethereum, Coinbase)
Portfolio Index Unit	Fractional exposure to a curated group of portfolio assets	Composite settlement

3.7 Due Diligence System

For high-value deals — particularly in the Financial Instruments Market and large Portfolio Commodity trades — kogi-exchange provides a structured Due Diligence (DD) framework that gives buyers access to governed, auditable portfolio intelligence before committing.

DD Component	Description
Portfolio Health Report	kogi-engine PortfolioHealthScore: overall score, signal breakdown, risk flags
Project Metrics	Velocity, completion rates, sprint data, milestone history
Financial Summary	Revenue, expenses, budget utilization, cash flow signals from kogi-bank
Audit Trail	Full immutable EventLog from the Portfolio System for the subject component
Risk Assessment	kogi-engine RiskEngine: risk score, risk optimization plan, red flags
Asset Valuation	AI-computed AssetValue model with comparable transactions
Artifact Maturity	ArtifactMaturity score: completeness, test coverage, documentation, update frequency
Governance Snapshot	Policy compliance, approval workflows, charter commitments
Social Proof	Community engagement, follower/subscriber count, review summary, spread analytics

DD reports are generated on-demand by the kogi-engine and presented in the Deal Room. They are version-snapshotted at the time of deal acceptance to create an immutable record of the basis on which the deal was agreed.

3.8 Third-Party Integrations

Category	Platforms	Integration Type
Investment / Trading	Robinhood, SoFi, Interactive Brokers	Portfolio linking, position sync
Crypto / Web3	Coinbase, Ethereum, Polygon, Solana	Token bridge, wallet connect, NFT integration
Banking	Stripe, Plaid, Chase, Wells Fargo	Account linking, payment rails via kogi-bank
Equity Platforms	AngelList, Republic, Wefunder	Round sync, cap table bridge
Commodity Markets	CME, commodity APIs	Real-world asset price feeds

4. SHARED INFRASTRUCTURE

4.1 Deal Room

The Deal Room (KRMS type=deal) is the negotiation and coordination space that every Marketplace and Exchange transaction moves through. It is a purpose-built Room from the kogi-community Rooms system, pre-loaded with deal context and equipped with AI assistance.

4.1.1 Deal Room Components

- Conversation thread — full message history between buyer and seller
- Proposal panel — active Proposal with version history and side-by-side comparison
- Document vault — attached files, contracts, DD reports, portfolio previews
- Milestone tracker — visual milestone board with status and escrow lock amounts
- Oba AI panel — AI assistant for drafting terms, comparing proposals, flagging risks
- Activity log — immutable deal event history (viewed, proposed, accepted, paid, delivered)
- Escrow status widget — live view of escrow balance and release conditions from kogi-bank
- Action bar — Propose, Accept, Counter, Fund Escrow, Deliver, Approve, Close

4.2 Escrow & Payment

Payment for Marketplace and Exchange transactions flows through kogi-bank's escrow subsystem. This ensures that funds are protected, conditions are enforced, and settlement is automated where possible.

4.2.1 Escrow Lifecycle

Stage	Description
Funded	Buyer deposits funds into escrow when deal is accepted. Funds locked in kogi-bank.
Milestone Release	Seller delivers milestone. Buyer approves. Partial escrow released to seller.
Full Release	Final deliverable approved. Remaining escrow balance released.
Dispute Hold	Either party raises a dispute. Funds frozen pending resolution.
Refund	Deal cancelled before delivery. Funds returned to buyer minus any completed milestones.
Auto-Release	Configurable: if buyer doesn't dispute within N days of delivery, funds auto-release.

4.3 Ratings & Reviews

Every completed deal generates a mutual review. Reviews are tied to both the listing and the worker's Portfolio Resource item, feeding the platform-wide reputation graph.

Field	Description
review.rating	1–5 star score
review.dimensions	Quality, Communication, Timeliness, Value (each 1–5)
review.text	Free-text review body
review.author_id	Reviewing party
review.subject_id	Reviewed party
review.deal_id	Linked deal reference
review.verified	Boolean: deal-linked reviews are auto-verified
review.ai_sentiment	kogi-engine sentiment analysis: positive neutral negative + score
review.response	Reviewed party's optional public response

4.4 Search & Discovery

Marketplace and Exchange discovery is powered by a multi-layer search and recommendation stack, drawing on the kogi-engine sub-engines:

Capability	Engine	Description	Trigger
Full-text Search	SearchEngine	Indexed listing titles, descriptions, tags, skills	User search query
Semantic Search	SearchEngine + QueryEngine	Intent-aware search via AI embedding similarity	User search query
Personalized Recommendations	RecommendationEngine	Collaborative + content-based filtering based on user profile	Feed refresh
Discover	AnalyticsEngine	AI-curated topic expansions beyond immediate interests	User action
Explore	AnalyticsEngine + GraphEngine	Deep expansion into related topics and entities	User action
AI Match	MatchEngine	Multi-dimensional worker-project-resource matching	Portfolio scan / blocker
Trending	TelemetryEngine	Listings with highest engagement velocity	Real-time feed

Similar Listings	RecommendationEngine	Content-based similarity on viewed listing	Listing view
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4.5 Notifications & Alerts

The Feed and Notification system surfaces Marketplace and Exchange events in real time:

- New proposal received on active listing
- Counterproposal submitted by buyer
- Deal accepted — escrow funded
- Milestone due or overdue
- Escrow released — payment received
- New review posted on your listing
- AI match found for an open requirement
- Campaign goal reached
- Bid matched on Exchange order
- Contract expiry approaching

5. AI INTEGRATION

5.1 kogi-engine Connections

The Marketplace and Exchange are among the most AI-intensive modules in the Kogi platform. Every sub-engine in kogi-engine has a meaningful role in powering the economic layer:

Sub-engine	Role in Marketplace / Exchange	Key Output	Trigger
MatchEngine	Multi-dimensional worker-to-project matching	Ranked match list with compatibility scores	Blocker detected / RFP posted
RecommendationEngine	Personalized listing discovery	Curated listing feed per user persona	Feed refresh
SearchEngine	Full-text + semantic listing search	Ranked search results with highlights	User search
RiskEngine	Deal and counterparty risk scoring	PortfolioHealthScore + risk flags	Deal initiation / DD request
AnalyticsEngine	Portfolio signal computation for pricing	AssetValue + ArtifactMaturity models	Listing creation / DD
PersonalizationEngine	Feed density, ordering, segment targeting	ContentDeliveryConfig	Session start
TelemetryEngine	Trending listings, market activity velocity	EngineFlowSnapshot with market signals	Continuous
QueryEngine	SQL optimization for deal + order queries	QueryPlan with optimized execution	Complex data queries
GraphEngine	Dependency impact analysis on deal subjects	ImpactReport for deal scope analysis	Deal due diligence
OptimizationEngine	Pricing and resource allocation recommendations	OptimizationPlan with lever suggestions	Campaign / pricing review

5.2 Oba AI Agent Workflows

The Oba AI assistant runs several proactive workflows specifically for Marketplace and Exchange activity:

5.2.1 Sprint Unblocking via Labor Market

When a project story is blocked, Oba automatically: (1) analyzes the blocker to build a skill profile, (2) searches the Labor Market for matching contractors via MatchEngine, (3) prepares

personalized inquiry messages with project context attached, and (4) presents a ready-to-send shortlist to the user for one-click approval.

5.2.2 Listing Quality Optimization

When a listing is underperforming (low CTR, low conversion), Oba analyzes the listing against high-performing comparables in the same category, surfaces specific improvement suggestions (title, description, pricing, tags), and offers to rewrite the listing body with one click.

5.2.3 Proposal Drafting & Review

When a buyer submits an inquiry, Oba drafts a tailored Proposal response using the seller's portfolio context, current availability from kogi-gig, and pricing intelligence from the Exchange. It flags unusual terms in received proposals and surfaces risk factors identified by the RiskEngine.

5.2.4 Deal Monitoring & Escrow Management

Oba monitors all active deals for milestone due dates, delivery confirmation delays, and communication gaps. It proactively drafts follow-up messages, flags overdue deliverables, and prepares escrow release or dispute documentation when needed.

5.2.5 Campaign Intelligence

For active fundraising or equity campaigns, Oba tracks contribution velocity against the goal deadline, identifies likely-to-convert prospects from the platform's social graph, and drafts targeted outreach messages for the campaign owner's approval.

5.2.6 Morning Portfolio Briefing — Market Section

The daily AI briefing includes a Marketplace & Exchange summary: active deals requiring attention, open proposals pending response, approaching contract expiries, escrow releases awaiting confirmation, and AI-matched opportunities that became available since the last session.

6. PORTFOLIO SYSTEM INTEGRATION

6.1 The Portfolio as the Source of Truth

Every entity traded on the Marketplace or Exchange is ultimately a Portfolio Component. The Portfolio System is the source of record for all listings, deals, orders, and their associated analytics. This means the full power of the Portfolio System — its audit trail, governance model, CRDT distribution, computational analytics, and permission hierarchy — is available to every economic transaction.

6.2 Integration Points

Integration Point	Direction	Description
Listing Creation	Portfolio → Market	Asset/Artifact/Resource with visibility=Public is surfaced to the Marketplace listing index
ActionKind::Invest	Market → Portfolio	Investor action recorded in Portfolio Component users.investors set
ActionKind::Donate	Market → Portfolio	Donor action recorded in Portfolio Component users.donors set
ActionKind::Subscribe	Market → Portfolio	Subscription recorded in analytics.subscribers; seller notified on updates
Deal Completion	Market → Portfolio	Completed deal creates a new linked Asset or Resource item in buyer's portfolio
Campaign Funding	Market → Portfolio	Campaign funding recorded via ResourceAllocation (Budget kind) in Portfolio System
Due Diligence	Portfolio → Market	PortfolioHealthScore, ArtifactMaturity, and EventLog exported to Deal Room DD panel
Milestone Tracking	Portfolio ↔ Market	Project sprint milestones linked to Deal milestone schedule
Asset Valuation	Portfolio → Market	AssetValue model consumed for listing price intelligence and DD reports
Version Notifications	Portfolio → Market	Portfolio Component version bump triggers subscriber notifications

6.3 Visibility & Access Control

The Portfolio System's permission and visibility model governs what can be listed and who can see it:

Visibility Level	Marketplace / Exchange Behavior
Private	Not surfaced to any marketplace or exchange. Owner only.
Protected	Visible only to explicitly invited users or community members. Invite-only listings.
Public	Fully discoverable on Marketplace and Exchange. Indexed by SearchEngine.
Community	Visible only to members of linked Community Space(s). Semi-public listings.

7. DATA ARCHITECTURE

7.1 Database Schema Design

7.1.1 Primary Storage — PostgreSQL

Marketplace and Exchange data is persisted in PostgreSQL with JSONB for flexible payload fields. Core tables:

Table	Description
listings	Primary listing records: id, type, owner_id, portfolio_component_id, price, status, metadata JSONB
orders	Exchange orders: id, market, type, subject JSONB, price JSONB, status, counterparty_id
proposals	Proposal records: id, listing_id, buyer_id, seller_id, scope, terms JSONB, milestones JSONB, status
deals	Active and completed deals: id, proposal_id, order_id, escrow_id, status, timeline JSONB
campaigns	Campaign records: id, type, goal JSONB, raised, backers JSONB, equity_terms JSONB, status
reviews	Review records: id, deal_id, author_id, subject_id, rating, dimensions JSONB, ai_sentiment
deal_rooms	Deal Room references: id, deal_id, room_id (FK to KRMS rooms)
escrow_locks	Escrow lock records (FK to kogi-bank): id, deal_id, amount, status, release_conditions JSONB
listing_analytics	Time-series analytics: listing_id, timestamp, impressions, clicks, conversions

7.1.2 Cache Layer — Redis

- Active listing index — hot-path read cache for search results and recommendation feeds
- Order book state — live bid/ask queues for Exchange markets
- Session deal context — current Deal Room state per active negotiation
- User match cache — MatchEngine result cache keyed by user profile hash

7.1.3 Event Streaming — Kafka

All Marketplace and Exchange events are published to Kafka for downstream consumption by kogi-engine and the Data Lake:

Topic	Events	Consumers
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marketplace.listings	listing.created, listing.updated, listing.published, listing.sold	SearchEngine, AnalyticsEngine, Feed Service
marketplace.deals	proposal.created, proposal.accepted, deal.started, deal.completed, deal.disputed	kogi-bank, AnalyticsEngine, AI Agent
exchange.orders	order.placed, order.matched, order.cancelled, order.expired	MatchEngine, AnalyticsEngine, Feed Service
marketplace.campaigns	campaign.created, campaign.backed, campaign.funded, campaign.failed	kogi-bank, AnalyticsEngine, kogi-organizations
marketplace.reviews	review.created, review.response_added	RecommendationEngine, SearchEngine

7.2 Analytics

The Marketplace and Exchange analytics system tracks performance across five dimensions, all powered by kogi-engine's AnalyticsEngine:

7.2.1 Listing Performance Metrics

- Impressions — total times listing was surfaced in search or feed
- Reach — unique users who viewed the listing
- Click-through Rate (CTR) — listing views / impressions
- Conversion Rate — proposals or orders / listing views
- Revenue — total value of completed deals originating from this listing
- Average Deal Value — mean deal size across all completed engagements

7.2.2 Worker / Seller Performance Metrics

- Completion Rate — completed deals / accepted deals
- On-Time Delivery Rate — delivered by milestone date / total milestones
- Average Rating — aggregate review score across all completed deals
- Response Time — median time to respond to new inquiries
- Repeat Buyer Rate — buyers who initiated more than one deal
- Revenue Growth Rate — month-over-month change in earned revenue

7.2.3 Market Health Metrics

- Liquidity — ratio of active orders to total registered users
- Match Rate — % of posted requirements that receive a qualified proposal
- Time to Match — median time from requirement posting to first qualified proposal
- Deal Close Rate — % of proposals that convert to funded deals
- Market GMV — Gross Merchandise Value of completed transactions per period

8. UX ARCHITECTURE & USER FLOWS

8.1 Marketplace UI Structure

The kogi-marketplace UI is organized around a primary discovery surface and a deal management layer:

View	Description
Marketplace Home	Personalized feed: AI-curated recommendations, trending listings, recent activity from followed workers
Browse / Search	Category-filtered search with facets: type, skills, price range, rating, availability, tags
Listing Detail	Full listing page: description, media, reviews, seller portfolio preview, Propose / Buy / Save actions
My Listings	Seller dashboard: all listings with performance analytics, edit, pause, promote controls
Proposals & Deals	Buyer/Seller inbox: active proposals, deal rooms, pending actions
Deal Room	Negotiation space: message thread, proposal panel, milestone tracker, escrow widget, Oba AI panel
Campaigns	Campaign browser: category filter, progress bars, backer counts, invest / donate actions
Saved / Bookmarks	User's saved listings collection
Analytics Dashboard	Seller performance: revenue, CTR, conversion, review trends, AI optimization suggestions

8.2 Exchange UI Structure

View	Description
Exchange Home	Market overview: active orders, trending trades, AI match alerts, portfolio signals
Labor Market	Worker discovery: skill search, availability filter, AI match recommendations, post a requirement
Commodities Market	Portfolio item browser: trade type filter, asset category, valuation range
Financial Market	Investment opportunities: equity rounds, revenue share offers, campaign listings, due diligence access
Resource Market	Available resources: compute, licenses, equipment, capital offerings
Order Book	Live bid/ask order book for active markets
My Orders	All open and completed orders with status tracking

Deal Room	Shared with Marketplace — same negotiation infrastructure
Due Diligence Portal	DD report viewer: portfolio health, risk score, audit trail, asset valuation

8.3 Core User Flows

8.3.1 Buyer — Find & Hire a Worker

1. Open Marketplace → Work category → search skills + availability
2. Browse AI-matched candidates or search results
3. View worker listing: portfolio preview, reviews, rate, AI reputation score
4. Open Deal Room via 'Propose' → write inquiry or let Oba draft it
5. Seller submits Proposal with scope, timeline, milestones, price
6. Review and negotiate terms in Deal Room
7. Accept Proposal → Fund Escrow via kogi-bank
8. Monitor work progress via milestone tracker
9. Approve deliverables → Escrow releases to worker
10. Submit mutual review → Analytics updated

8.3.2 Seller — List and Sell an Asset

11. Open kogi-portfolio → select Asset or Artifact → Set visibility=Public
12. Marketplace listing wizard auto-populated from Portfolio Component data
13. Set price, pricing model, tags, and media → Publish
14. Receive notifications as buyers save, view, and propose
15. Respond to proposals via Deal Room (Oba can draft responses)
16. Accept → Escrow funded → Deliver asset or access
17. Completion confirmed → Payment received → Review collected

8.3.3 Exchange — Post a Requirement (RFP)

18. Open Exchange → Labor Market → Post Requirement
19. Describe scope, required skills, timeline, budget ceiling
20. AI MatchEngine immediately surfaces pre-qualified candidates
21. Sellers submit competing proposals
22. Compare proposals (Oba can summarize and compare)
23. Select and accept best proposal → Deal Room activated
24. Fund escrow → Work begins

8.3.4 Campaign — Launch Equity Round

25. Open Marketplace → Campaigns → Create Campaign
26. Link to Portfolio Program or Project
27. Set campaign type (Equity), funding goal, equity terms, deadline
28. AI generates campaign description draft from portfolio context
29. Publish → Share to Community via Spaces and Feed

- 30. Track contributions in real-time; Oba monitors velocity
- 31. Goal reached → Funds distributed via kogi-bank → Cap table updated in kogi-organizations

9. TECHNICAL ARCHITECTURE

9.1 Service Architecture

Service	Description
kogi-marketplace (Go)	Primary REST/GraphQL API service for listings, proposals, and campaigns. Publishes events to Kafka.
kogi-exchange (Go)	Order book management, trading mechanics, and market data API. Publishes events to Kafka.
kogi-deal (Go)	Deal lifecycle orchestration: proposal → deal → escrow → delivery → settlement → review.
kogi-match (Go bridge)	Thin bridge service wrapping MatchEngine gRPC calls from kogi-engine Scala service.
kogi-engine (Scala)	Intelligence substrate: MatchEngine, RecommendationEngine, RiskEngine, SearchEngine, AnalyticsEngine.
kogi-bank (Go/Rust)	Escrow, payment, wallet, settlement, invoice services consumed by both Market and Exchange.
kogi-network (Go)	Gateway management, publish/subscribe, service-to-service communication.

9.2 Data Flow

User Action → kogi-network (Gateway) → kogi-marketplace / kogi-exchange API → Portfolio System (Rust) → Kafka Event Bus → kogi-engine (Scala) → Recommendations / Risk / Match → Feed Service → Client Notification

9.3 Technology Stack

Layer	Technology	Component	Notes
Core Services	Go	kogi-marketplace, kogi-exchange, kogi-deal	REST + gRPC; Kafka producers
Systems Layer	Rust	PortfolioSystem, core modules	In-memory + PostgreSQL backend planned
Intelligence	Scala 3 / JVM	kogi-engine (all sub-engines)	gRPC server on port 9100
Database	PostgreSQL	Primary persistence	JSONB for flexible payloads

Cache	Redis	Hot-path reads, order book, session state	Pub/sub for real-time fanout
Streaming	Apache Kafka	Event bus, analytics pipeline, audit log	Topics per module
Web Client	Angular + TypeScript	kogi-web-client	WebSocket for real-time Deal Rooms
Mobile	Kotlin / iOS	kogi-mobile-client	Push notifications for deal events
Search Index	Meilisearch / Typesense	Listing search backend	Fed from Portfolio plugin hooks
Build	Bazel	Monorepo build system	Cross-language build consistency

9.4 Security & Governance

- All marketplace and exchange actions validated against PortfolioSystem PermissionTier hierarchy
- Deal escrow operations require explicit user approval — no AI-initiated fund movements
- Full audit trail via Portfolio EventLog for every listing, deal, order, and payment action
- PolicyEngine hooks enforced on all listing publications and deal completions
- Federated identity via OIDC/SAML for cross-platform authentication
- Encrypted Deal Room messages; at-rest encryption for DD report snapshots
- Rate limiting and anomaly detection on order book to prevent market manipulation

10. OPEN ITEMS & FUTURE CONSIDERATIONS

10.1 Near-Term Open Items

Item	Description
Order Book Persistence	Exchange order book currently in-memory. Needs Redis-backed persistent queue with Kafka replay for crash recovery.
Escrow Smart Contracts	For crypto/token deals, escrow should be backed by on-chain smart contracts. Define bridge protocol between kogi-bank and supported chains (Ethereum, Polygon).
AI Writeback Protocol	Define typed protocol for Oba agent to create Proposal drafts, update deal metadata, and append governance notes without bypassing PermissionTier checks.
Match Algorithm Tuning	MatchEngine currently uses linear multi-dimensional scoring. Define weighting schema per market type (Labor vs. Commodities vs. Financial) with A/B experiment infrastructure.
CRDT for Deal State	Deal state (proposal counters, milestone status) needs CRDT-safe merge strategy for multi-device and offline scenarios.
SearchEngine Warm-up	SearchEngine listing index must be populated via Portfolio System plugin hooks on component creation/update. Wire <code>on_component_created</code> → listing index upsert.
Provider Integration Depth	Initial provider integrations (Upwork, Fiverr, etc.) are tile-level deep links. Full API integrations for profile import and order sync require provider-specific OAuth flows.

10.2 Phase 4 Considerations

- Global marketplace liquidity pools — pooled order routing across federated Kogi nodes
- Autonomous agent-driven deal negotiation — Oba negotiates within user-defined constraints without per-action approval
- On-chain settlement for financial instruments — token issuance, distribution, and cap table management on-chain
- Portfolio Index Products — curated composite indices of portfolio assets tradable as a single instrument
- Cross-platform marketplace federation — federated listing discovery across ShangoOS platforms (Kogi, Ume, Qala)
- Algorithmic pricing suggestions — ML-driven price optimization based on market comps, demand signals, and portfolio health
- DAO governance integration — on-chain voting for community fund distributions and cooperative economic decisions

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